

$$1.b. \text{ NKG. } H \leq G \Rightarrow HN = G.$$

Def NKG $H \leq G$ $H \cap N = \{e\}$ Then

HN is a semidirect product.

Further: $H \trianglelefteq G$. Semidirect is called direct product.

G is abelian $H_1, H_2 \leq G$ $H_1 \cap H_2 = \{e\}$.

$H_1 + H_2$ is a direct sum.

$$HN := H \ltimes N \quad (H \times N)$$

"is isomorphic"

$$\text{prop: } ① HN = H \ltimes N$$

$$② e = hn \Rightarrow h = n = e$$

$$③ h_1 n_1 = h_2 n_2 \Rightarrow h_1 = h_2 \quad n_1 = n_2$$

$$④ HN/N \cong H \Leftrightarrow HN \xrightarrow{\pi} H \times N = \{(h, n) \mid h \in H, n \in N\}$$

$$HN \leq G$$

$$h_1 n_1 \cdot h_2 n_2 = h_1 h_2 (h_2^{-1} n_1 h_2) n_2$$

$\downarrow \quad \quad \downarrow$
 $H \quad \quad N$

$$① I_{h_2} \in \text{Aut}(N).$$

$$② \varphi: H \rightarrow \text{Aut}(N)$$

$H \rightarrow I_h$ hom.

$$(h_1 n_1)(h_2 n_2) = h_1 h_2 (\varphi(h_2)(n_1) \cdot n_2)$$

① suppose H and N are groups

$\varphi: H \rightarrow \text{Aut}(N)$ is a homomorphism

$$H \ltimes N \text{ define } (h_1, n_1)(h_2, n_2) := (h_1 h_2, \varphi(h_2)(n_1) n_2).$$

Then $H \ltimes N$ is a group

$$① ((h_1, n_1)(h_2, n_2))(h_3, n_3) = (h_1 h_2, \varphi(h_2)(n_1) n_2)(h_3, n_3)$$

$$② (e_H, e_N)$$

$$(h, n)(e_H, e_N) = (h e_H, \varphi(e_H)(n) \cdot e_N) = (h, n)$$

$$(e_H, e_N)(h, n) = (e_H h, \varphi(h)(e_N) n) = (h, n)$$

$$\varphi(h) \in \text{Aut}(N)$$

$$(h, n)(h', n') = (e_H, e_N)$$

$$(h h', \varphi(h')(n) \cdot n')$$

$$h h' = e_H \Rightarrow h = h^{-1}$$

$$\varphi(h')(n) n' = e.$$

$$n' = (\varphi(h')(n))^{-1} = (\varphi(h^{-1})(n))^{-1}.$$

$\therefore H \ltimes N$ is a group

denoted by $H \ltimes N$

$$(e_H, e_N) \trianglelefteq H \ltimes N.$$

$$(H, e_N) \leq H \ltimes N.$$

$$(H, e_N) \cap (e_H, N) = \{(e_H, e_N)\}.$$

$$H \ltimes N = (H, e_N) \ltimes (e_H, N) := \underline{H \ltimes N}.$$

external semidirect product $\quad \quad \quad$ internal semidirect product.

$$H \ltimes N \xrightarrow{\pi} H \times N.$$

$$\varphi: H \rightarrow \text{Aut}(N).$$

$$h \mapsto I_h.$$

$$\text{Example. } S_3 \supseteq \langle (123) \rangle.$$

$$[S_3: \langle (123) \rangle] = 2 = \frac{|S_3|}{|\langle (123) \rangle|} = \frac{6}{3} = 2.$$

$$\therefore \langle (123) \rangle \trianglelefteq S_3.$$

$$\langle (12) \rangle \cap \langle (123) \rangle = \{(1)\}.$$

$$S_3 = \langle (12) \rangle \ltimes \langle (123) \rangle$$

$$\text{property. } H_1 \trianglelefteq G, H_1 \cap H_2 = \{e\}.$$

$$H_1 \times H_2$$

$$\forall h_1 \in H_1, h_2 \in H_2$$

$$h_1 h_2 = h_2 h_1$$

$$\Leftrightarrow h_1 h_2 (h_2 h_1)^{-1} = e$$

$$h_1 h_2 h_2^{-1} h_1^{-1} = (h_1 h_2 h_1^{-1}) h_2 \in H_2$$

$$= h_1 (h_2 h_1^{-1} h_2^{-1}) \in H_1$$

$$\therefore h_1 h_2 h_1^{-1} h_2^{-1} \in H_1 \cap H_2 = \{e\}$$

Cor. H_i abelian. so $H_1 \times H_2$

$$|G| = 77. \quad K_7 = 1 \Leftrightarrow P \triangleleft G.$$

$$K_{11} = 1 \Leftrightarrow Q \triangleleft G.$$

$$P \cap Q = \{e\}$$

$$G = P \times Q \quad \therefore G \text{ is abelian.}$$

semidirect product. internal $\simeq$ external.

$\downarrow$

direct product.

internal $H_1 \times H_2$

$$\uparrow 1:1$$

$$\{(h_1, h_2) \mid h_i \in H_i\}$$

$$(h_1, h_2)(h'_1, h'_2) = h_1(h_2 h'_1)h'_2$$

$$= (h_1 h'_1)(h_2 h'_2)$$

external direct product.

$$\frac{H_1}{H_2} \Rightarrow H_1 \times H_2 = \{(h_1, h_2) \mid h_i \in H_i\}$$

$$(h_1, h_2)(h'_1, h'_2) = (h_1 h'_1, h_2 h'_2)$$

$$= (H_1, e_{H_2}) \times (e_{H_1}, H_2)$$

$$\begin{matrix} \text{SII} \\ H_1 \end{matrix} \quad \begin{matrix} \text{SII} \\ H_2 \end{matrix}$$

Definition,

suppose $\{G_i \mid i \in I\}$ is a family of groups.

$$\prod_{i \in I} G_i = \{(a_i)_{i \in I} \mid a_i \in G_i, \forall i \in I\}$$

$$\Leftrightarrow \{f: I \rightarrow \bigcup_{i \in I} G_i \mid f(i) = a_i \in G_i\}$$

$$(a_i)_{i \in I} (b_i)_{i \in I} = (a_i b_i)_{i \in I}$$

direct product of $\{G_i \mid i \in I\}$

$$I = \{1, 2, \dots, n\}$$

$$\prod_{i \in I} G_i := \prod_{i=1}^n G_i \text{ (or } G_1 \times G_2 \times \dots \times G_n)$$

$$= \{(a_1, a_2, \dots, a_n) \mid a_i \in G_i\}$$

$$(a_1, a_2, \dots, a_n)(b_1, b_2, \dots, b_n) = (a_1 b_1, a_2 b_2, \dots, a_n b_n)$$

Example $G_i = G, \forall i \in I$,

$$\prod_{i \in I} G_i := G^I$$

$$= \{f: I \rightarrow G\}$$

$$G = (\mathbb{R}, +), I = \mathbb{N}$$

$$\mathbb{R}^{\mathbb{N}} = \{(a_i)_{i \in \mathbb{N}} \mid a_i \in \mathbb{R}\}$$

$$= \{\{a_n\}_{n \in \mathbb{N}} \mid a_n \in \mathbb{R}\}$$

$$\{a_n\} + \{b_n\} = \{a_n + b_n\}$$

weak direct product.

$$\text{Def } \{G_i \mid i \in I\} \Rightarrow \prod_{i \in I} G_i$$

$$\prod_{i \in I}^w G_i = \{(a_i)_{i \in I} \mid a_i \in G_i, a_i = e_i \text{ for all but finite } i\}$$

direct sum: suppose $\{G_i \mid i \in I\}$ abelian group

$$\prod_{i \in I}^w G_i = \bigoplus_{i \in I} G_i$$

is called a direct sum of $\{G_i \mid i \in I\}$

Prop 1.6.1.

$$\prod_{i \in I} G_i \xrightarrow{\pi_i} G_i$$

$$\{a_i\}_{i \in I} \mapsto a_i$$

π_i are epimorphism

$$\textcircled{2} \prod_{i \in I} G_i \xrightarrow{f_i} G_i$$

$$\exists! \psi \uparrow f_i \text{ (hom)}$$

$$\text{i.e. } \pi_i \circ \psi = f_i \quad \forall i \in I$$

$$\psi(h) := (f_i(h))_{i \in I}$$

$$\psi(h_1, h_2) = (f_i(h_1, h_2))_{i \in I}$$

$$= (f_i(h_1) f_i(h_2))_{i \in I}$$

$$= \psi(h_1) \psi(h_2)$$

$$\pi_i \circ \psi(h) = \pi_i((f_j(h))_{j \in I}) = f_i(h)$$

$$\varphi: \pi_i \circ \varphi = f_i$$

$$\Rightarrow \varphi = \psi \Leftrightarrow \varphi(h) = \psi(h)$$

$$\varphi(h) = (a_i)_{i \in I} \Leftrightarrow a_i = f_i(h)$$

$$\pi_i \circ \varphi(h) = a_i = f_i(h)$$

$\psi: \bigtriangleup_{i \in I} f_i$
 suppose G is a group

$$\pi_i': G \rightarrow G_i$$

$$\begin{array}{ccc} & \uparrow & \nearrow \\ \exists \psi & & \forall f_i (i \in I) \\ \forall H. & & \end{array}$$

Thm. $G \cong \prod_{i \in I} G_i$